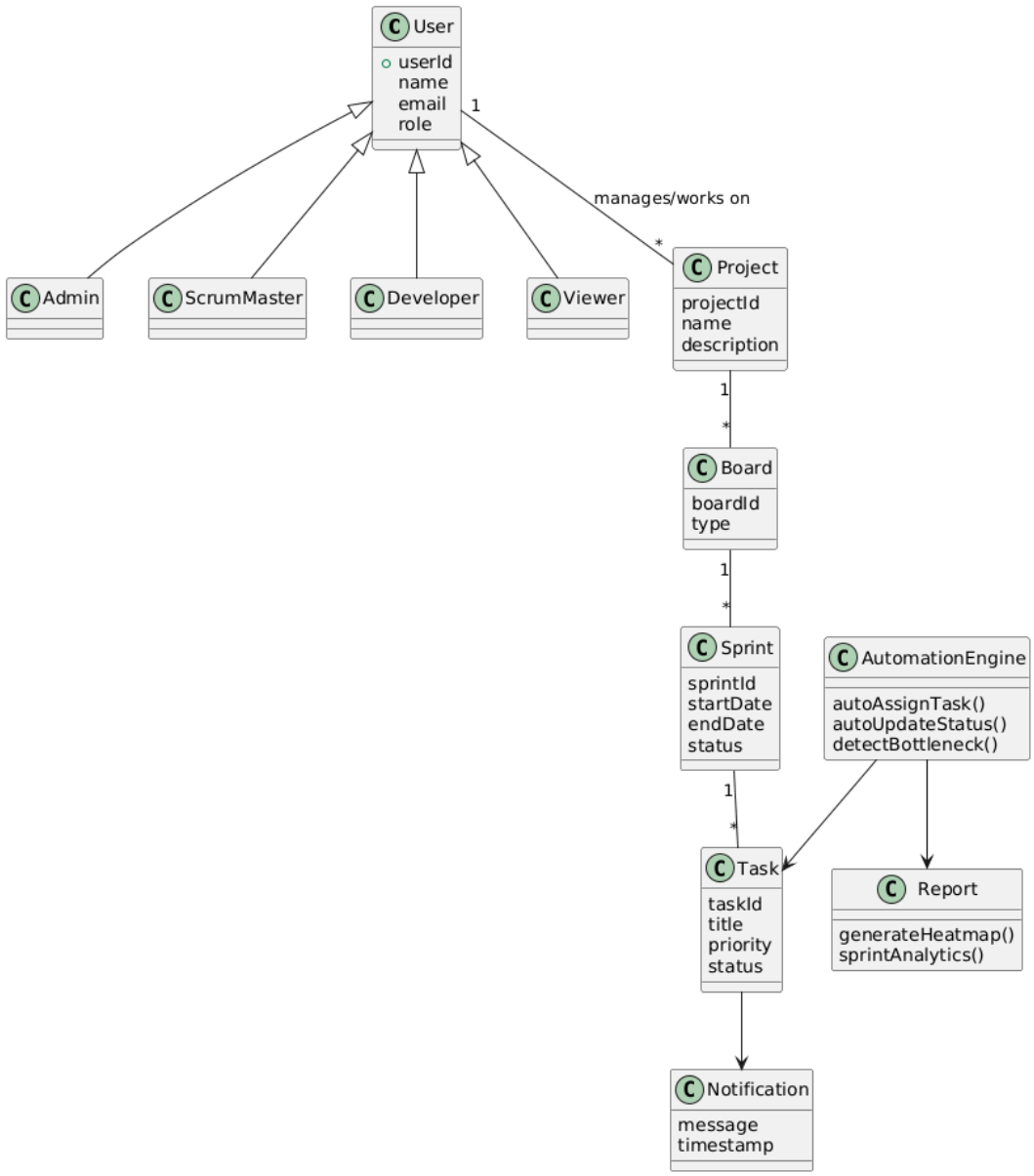
UML DIAGRAMS

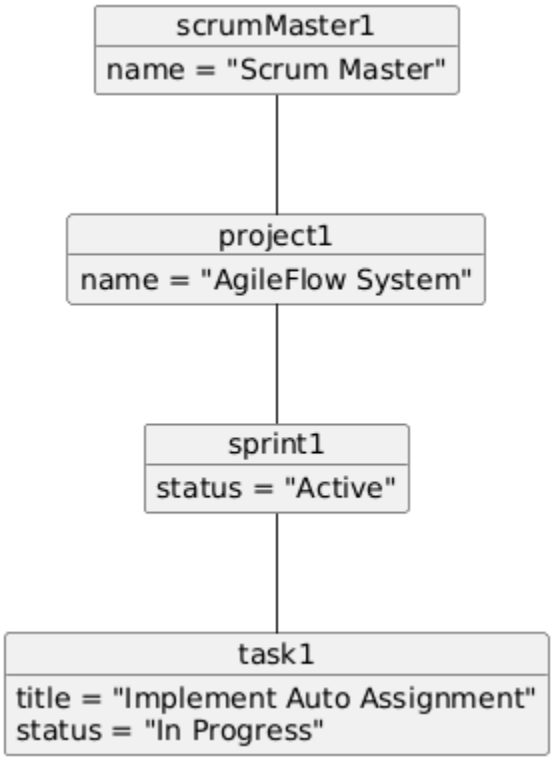
Class Diagram

Shows the system's classes (User, Role, Project, Board, Sprint, Task, Automation Rule, Notification, Report) with attributes and their relationships—useful for database/table design and object structure.



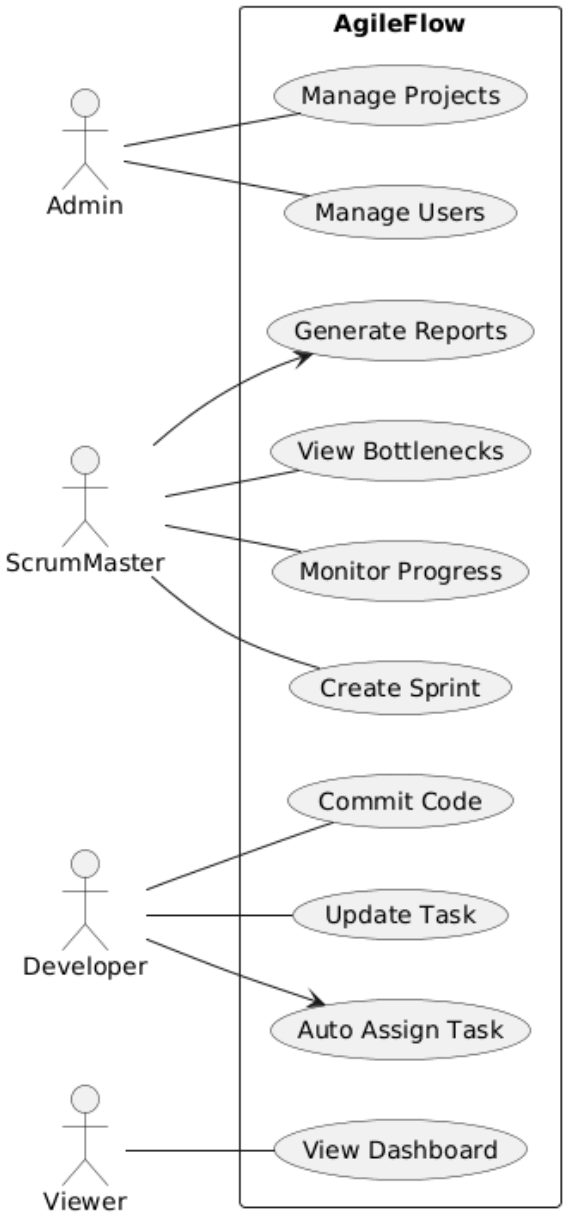
Object Diagram

Snapshot of runtime instances (example: a Scrum Master user, a project, sprint, and a task) illustrating how objects relate at a specific moment.



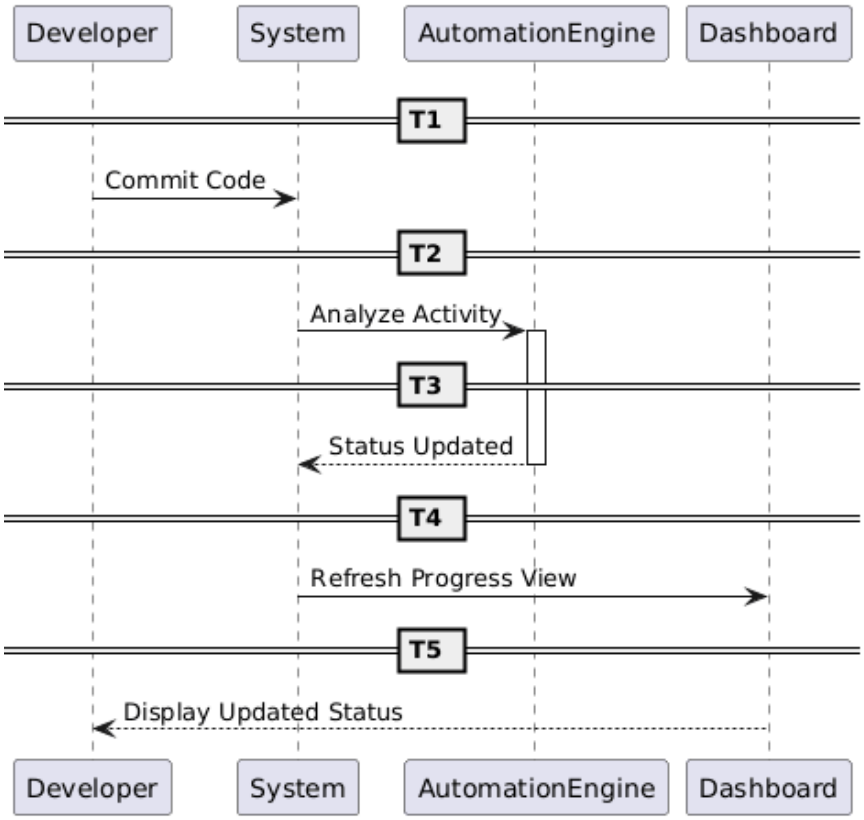
Use Case Diagram

Displays actors (Admin, Scrum Master, Developer, Viewer) and their high-level interactions with system features like managing users, creating projects, planning sprints, and viewing reports.



Sequence Diagram

Shows the interaction flow when creating a sprint — Scrum Master requests sprint creation, System validates and assigns tasks, and Developers receive notifications.

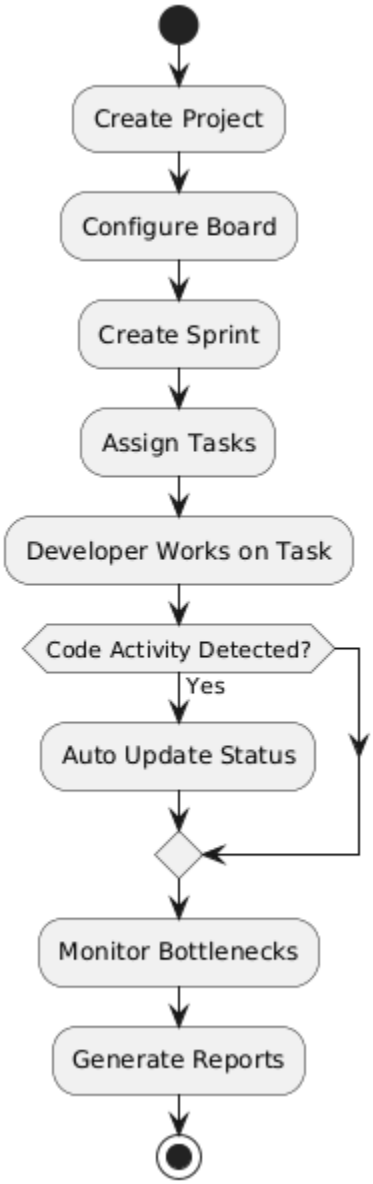


Collaboration Diagram

Visualizes components collaborating for task auto-assignment: Repo Hook -> System -> Automation Engine -> Task Service -> Notification Service -> Developer.

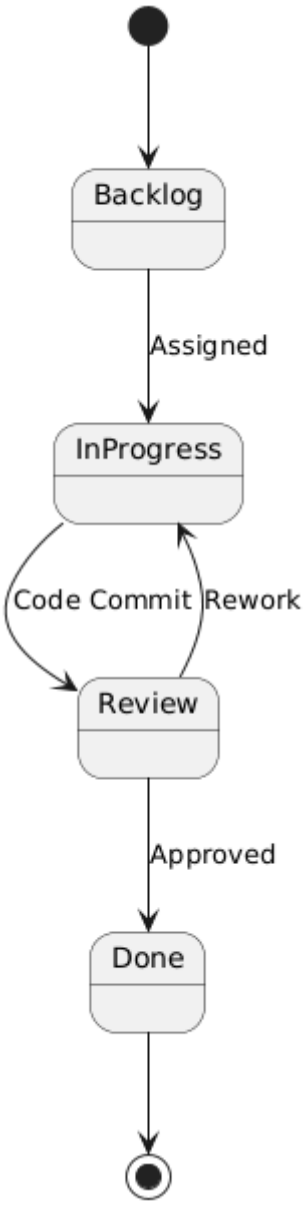
Activity Diagram

Represents the workflow for creating a sprint: enter details, validate, create record, and notify the team — with a loop back on validation failure.



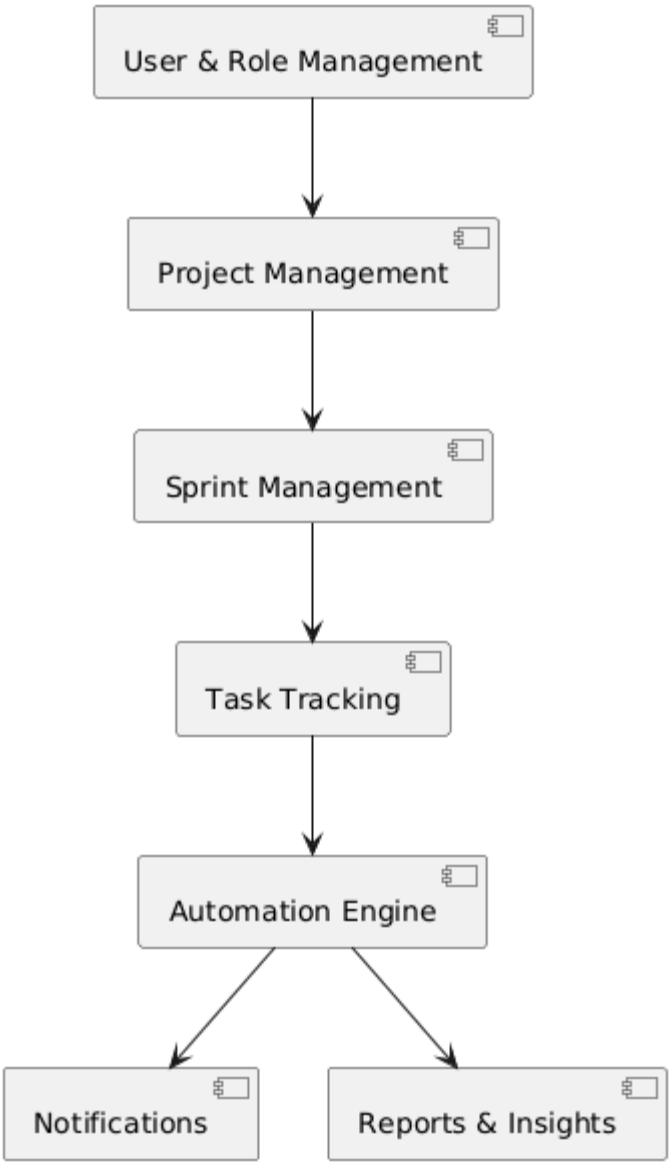
State Chart Diagram

Task lifecycle states (Backlog → ToDo → InProgress → InReview → Done) including transitions like start, submit, approve, and rework.



Component Diagram

High-level system components (Web Client, API, Auth Service, Automation Engine, Database, Repo Hook) and their interactions (REST, auth, webhooks).



Deployment Diagram

Runtime deployment topology showing Browser → Load Balancer → App Server (API, Automation Engine) → Auth Server and Database Server; Repo Server triggers webhooks.

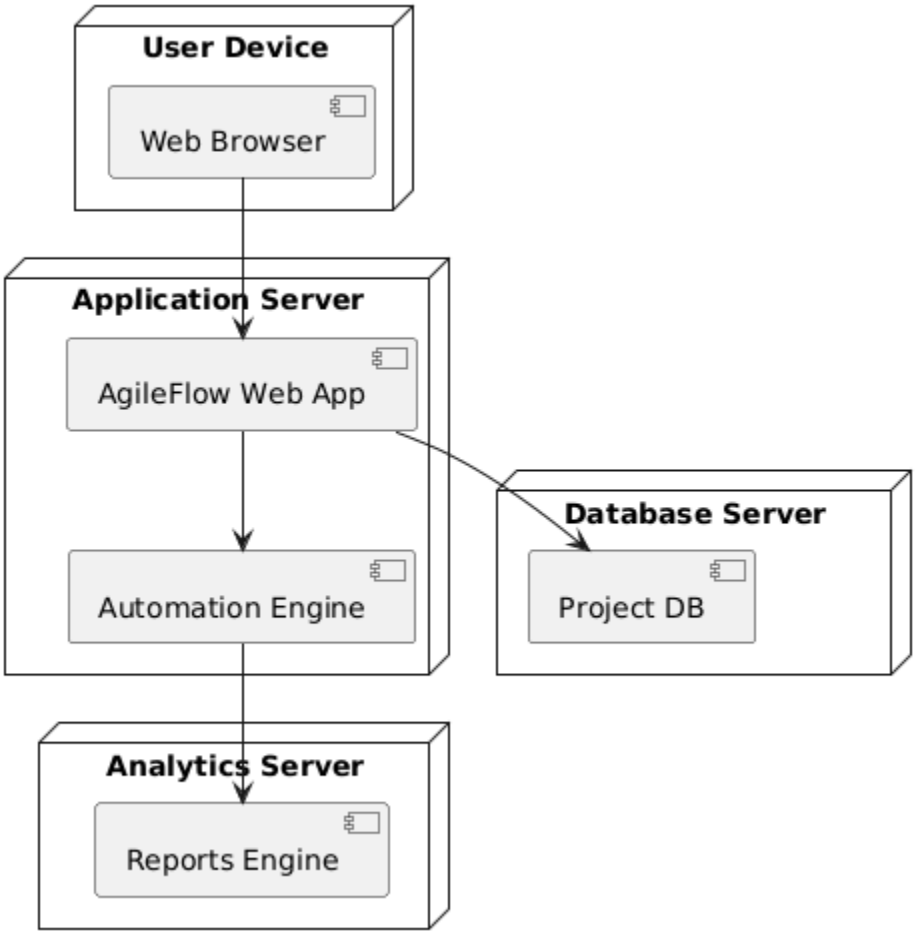


Table Design

Roles Table

Primary key: role_id

Column	Type	Description
role_id	BIGINT PK	Unique role identifier
name	VARCHAR(50) UNIQUE NOT NULL	Role name (Admin, Developer, etc.)

Users Table

Primary key: user_id

Foreign key: role_id

Column	Type	Description
user_id	BIGINT PK	Unique user
username	VARCHAR(100) UNIQUE NOT NULL	Login name
email	VARCHAR(255) UNIQUE NOT NULL	Email ID
password_hash	VARCHAR(255) NOT NULL	Hashed password
role_id	BIGINT FK	References roles(role_id)
created_at	TIMESTAMP	Creation timestamp

Projects Table

Primary key: project_id

Foreign key: user_id

Column	Type	Description
project_id	BIGINT PK	Unique project ID
name	VARCHAR(255) NOT NULL	Project name
owner_id	BIGINT FK	References users(user_id)
created_at	TIMESTAMP	Creation timestamp

Boards Table

Primary key: board_id

Foreign key: project_id

Column	Type	Description
board_id	BIGINT PK	Unique board ID
project_id	BIGINT FK	References projects(project_id)
name	VARCHAR(100) NOT NULL	Board name

Sprints Table

Primary key: sprint_id

Foreign key: project_id

Column	Type	Description
sprint_id	BIGINT PK	Unique sprint ID
project_id	BIGINT FK	References projects(project_id)
name	VARCHAR(100) NOT NULL	Sprint name
start_date	DATE NOT NULL	Sprint start date
end_date	DATE NOT NULL	Sprint end date

Automation Rules Table

Primary key: rule_id

Foreign key: project_id

Column	Type	Description
rule_id	BIGINT PK	Unique rule ID
project_id	BIGINT FK	References projects(project_id)
trigger	VARCHAR(255) NOT NULL	Automation trigger
action	VARCHAR(255) NOT NULL	Automation action

Tasks Table

Primary key: task_id

Foreign key: board_id, sprint_id, user_id

Column	Type	Description
task_id	BIGINT PK	Unique task ID
board_id	BIGINT FK	References boards(board_id)
sprint_id	BIGINT FK NULLABLE	References sprints(sprint_id)
title	VARCHAR(255) NOT NULL	Task title
description	TEXT	Task description
assignee_id	BIGINT FK	References users(user_id)
status	VARCHAR(50) NOT NULL	Task status
priority	VARCHAR(50) NOT NULL	Task priority

Notifications Table

Primary key: notification_id

Foreign key: user_id

Column	Type	Description
notification_id	BIGINT PK	Unique notification ID
user_id	BIGINT FK	References users(user_id)
message	TEXT NOT NULL	Notification message
is_read	BOOLEAN DEFAULT FALSE	Read status

Reports Table

Primary key: report_id

Foreign key: project_id

Column	Type	Description
report_id	BIGINT PK	Unique report ID
project_id	BIGINT FK	References projects(project_id)
report_type	VARCHAR(100) NOT NULL	Type of report
generated_at	TIMESTAMP	Generation timestamp